

BDM Capstone Project — Final Submission

Ramu Doors Private Limited

Data-Driven Sales and Inventory Optimization
for a B2B Door Manufacturing Enterprise

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1. Executive Summary

(Weightage: 15 marks)

Ramu Doors Private Limited is a B2B door manufacturing company based in Bengaluru, Karnataka, specializing in UPVC doors, FRP/GRP doors, flush doors, and premium solid doors for residential, commercial, and industrial applications. Founded in 2005 as Ramu Enterprises, the company has since transitioned into a Private Limited entity, currently employing 25–40 personnel across manufacturing, sales, and administration.

This project undertakes a comprehensive data-driven analysis to address three critical business challenges faced by the organization: (1) declining and fluctuating sales revenue due to seasonal demand cycles and competitive pressures, (2) the need for strategic market expansion into untapped geographic regions beyond the dominant Bangalore market, and (3) inefficient inventory management leading to poor availability of high-demand products and excessive stocking of slow-moving items.

The analysis was conducted using primary data collected directly from the company — comprising 336 sales transactions and a complete purchase register spanning August to October 2025, with a total 3-month revenue of approximately Rs. 4.10 Crore. Analytical techniques including time-series trend analysis, geographic and client segmentation, Pareto (80/20) analysis, and ABC inventory classification were applied using Microsoft Excel and Python.

Key findings reveal that September 2025 generated the highest monthly revenue (Rs. 1.90 Crore), Bangalore accounts for 62.7% of total revenue indicating market saturation risk, and just 3 out of 5 product categories (FRP/GRP Doors, UPVC Windows, Door Frames) contribute 84.3% of total revenue, classifying them as Class A inventory items. Based on these findings, actionable recommendations have been provided for demand planning, geographic diversification, client portfolio expansion, and inventory optimization to support the company's sustainable growth.

2. Organization Background

Ramu Doors Private Limited is a for-profit B2B manufacturing company based in Bengaluru, Karnataka, engaged in the production of UPVC doors, flush doors, FRP/GRP doors, and premium solid doors for residential, commercial, and industrial applications. The company was founded in 2005 as Ramu Enterprises by Mr. Ramakrishna M.C. and has since grown steadily by providing customized door and interior solutions to builders, contractors, and interior designers across Karnataka.

The organization operates as a small-to-medium-scale enterprise, with approximately 25–40 employees across manufacturing, sales, and administrative functions. It follows standard business operations for six days a week and maintains in-house facilities for fabrication, finishing, and installation. In recent years, the company transitioned into a Private Limited entity, reflecting its intent to formalize operations and support future expansion.

The company caters to two primary client segments — Builders (private construction firms) and Government/Institutional bodies — across multiple cities in Karnataka including Bangalore, Mysuru, Hubli, Mangaluru, and surrounding regions. Its product portfolio comprises five major categories: FRP/GRP Doors, UPVC Windows, Door Frames (FRP), Hardware & Fittings, and Modular Kitchens.

3. Problem Statement

PS-1: Sales Demand Fluctuation & Revenue Cyclicalities

The company is experiencing significant fluctuations in monthly sales revenue, with demand varying substantially across months. Understanding the nature and drivers of these fluctuations is critical for production planning, workforce management, and financial forecasting.

PS-2: Market Expansion into Untapped Regions

The company's revenue is heavily concentrated in Bangalore. The owner plans to expand the business into untapped markets in secondary cities but requires data-driven insights to evaluate the feasibility, potential, and strategic priorities for geographic and client segment diversification.

PS-3: Inventory Optimization & Product Availability

The company struggles with poor product availability due to inefficient inventory management. Popular products are often out of stock, while slow-moving items occupy inventory space. The owner seeks a data-driven approach to classify inventory, prioritize procurement, and optimize stock levels.

4. Detailed Explanation of Analysis Process/Method

(Weightage: 15 marks)

4.1 Data Collection and Sources

a) Sales Data:

Source: Collected directly from the business owner, Mr. Ramakrishna M.C., during personal visits to the firm

Period Covered: August 2025 – October 2025 (3 months)

Total Records: 336 sales transactions

Attributes: Invoice_ID, Date, Product_Type, Client_Name, Region, Client_Segment, Quantity, Unit_Price_Rs, Sales_Amount_Rs

b) Purchase Data:

Source: Extracted directly from the company's Tally accounting software

Period Covered: August 2025 – October 2025 (3 months)

Total Purchase Value: Rs. 2,01,43,596.75

Attributes: Date, Supplier_Name, Voucher_Type, Voucher_No, Debit_Amount_Rs

4.2 Metadata Summary

Sales Data Metadata:

Attribute	Description	Data Type
Invoice_ID	Unique invoice identifier (RD0001–RD0336)	Categorical / Nominal
Date	Date of sale transaction	Temporal
Product_Type	Category of product sold (5 categories)	Categorical / Nominal
Client_Name	Name of purchasing company/individual	Categorical / Nominal
Region	City location of the client	Categorical / Nominal
Client_Segment	Builder or Govt/Institutional	Categorical / Nominal
Quantity	Number of units sold per invoice	Numerical / Discrete
Unit_Price_Rs	Price charged per single unit (Rs.)	Numerical / Continuous
Sales_Amount_Rs	Total monetary value of the invoice (Rs.)	Numerical / Continuous

Purchase Data Metadata:

Attribute	Description	Data Type
Date	Date of purchase recorded	Temporal
Supplier_Name	Name of vendor/supplier	Categorical / Nominal
Voucher_Type	Internal classification (Purchase entries)	Categorical / Nominal
Voucher_No	Internal reference number	Categorical / Nominal
Debit_Amount_Rs	Total amount paid to supplier (Rs.)	Numerical / Continuous

4.3 Data Preparation

Before conducting the analysis, both datasets were imported into Microsoft Excel. The following data cleaning steps were performed: ensured uniform date formatting across all records, verified no duplicate invoice entries existed, handled null/missing values, cross-verified data consistency between sales and purchase records, and standardized categorical labels.

4.4 Descriptive Statistics

Statistic	Quantity	Unit_Price_Rs	Sales_Amount_Rs
Count	336	336	336
Mean	18.65	11,230	1,22,174
Std Dev	10.52	17,188	86,923
Min	1	549	12,516
25th %ile	10	3,499	56,195
Median	17	5,177	99,633
75th %ile	25	12,518	1,59,616
Max	50	1,08,089	4,44,843

Quantity: The mean order size is 18.65 units per invoice with moderate variability (SD = 10.52), typical of B2B order patterns. Unit Price: The mean (Rs. 11,230) is significantly higher than the median (Rs. 5,177), indicating a right-skewed distribution. Sales Amount: The mean invoice value is Rs. 1,22,174, accumulating to approximately Rs. 4.10 Crore over 3 months.

4.5 Analytical Techniques

Problem Statement	Method	Justification
PS-1: Sales Fluctuation	Time-series trend analysis	Identifies seasonal patterns and demand cyclicity
PS-2: Market Expansion	Geographic & client segmentation, AOV comparison	Quantifies concentration risk, identifies high-potential markets
PS-3: Inventory	Pareto Analysis, ABC Classification	Prioritizes high-value products for procurement focus

5. Results and Findings

(Weightage: 40 marks)

5.1 PS-1: Sales Demand Fluctuation Analysis

Monthly Revenue Comparison:

Month	Total Revenue (Rs.)	Transactions	Avg Unit Price (Rs.)	Revenue Share
August 2025	1,05,00,000	~105	10,398	25.6%
September 2025	1,90,00,000	~130	14,270	46.3%
October 2025	1,14,00,000	~101	8,323	27.8%
Total	~4,10,00,000	336	11,230	100%

Key Findings — PS-1:

1. September Dominance: September 2025 was the strongest performing month, generating Rs. 1.90 Crore in revenue — nearly 81% higher than August and 67% higher than October.
2. Demand Cyclicity: The data reveals a clear inverted-V pattern: demand ramps up from August, peaks in September, and contracts in October. This is likely driven by construction project timelines and festival-season procurement.
3. Pricing Variability: The average unit price dropped from Rs. 14,270 in September to Rs. 8,323 in October, indicating high-value project orders were concentrated in September.
4. Revenue Contraction Risk: The significant revenue drop in August and October highlights vulnerability to demand cycles, necessitating proactive demand forecasting.

[Insert Chart: Monthly Revenue Bar Chart here]

5.2 PS-2: Market Expansion Analysis

Revenue Distribution by Region:

Region	Revenue (Rs.)	Share (%)	Avg Order Value (Rs.)
Bangalore	~2,57,00,000	62.7%	1,15,000
Mysuru	~56,00,000	13.7%	1,40,485
Hubli	~48,00,000	11.7%	1,10,000
Mangaluru	~32,00,000	7.8%	95,000
Others	~17,00,000	4.1%	85,000

Key Findings — PS-2:

1. Bangalore Concentration Risk: 62.7% of total revenue is concentrated in Bangalore alone, exposing the company to localized competition and market saturation risks.
2. Mysuru — Highest AOV: Despite fewer clients, Mysuru yielded the highest average order value (Rs. 1,40,485), indicating significant untapped potential.
3. Hubli — Growth Potential: Hubli contributes 11.7% with a healthy AOV of Rs. 1,10,000 and established clients demonstrating existing market acceptance.

Revenue by Client Segment:

Segment	Revenue Share (%)	Avg Order Value (Rs.)	Transaction Share (%)
Builder	87.5%	~1,10,000	87.5%
Govt/Institutional	12.5%	~1,44,100	12.5%

Government/Institutional clients show a 31% higher average order value compared to Builder clients, yet represent only 12.5% of transactions — a significant untapped revenue opportunity.

[Insert Chart: Revenue Pie Chart by Region here]

5.3 PS-3: Inventory Optimization — Pareto & ABC Analysis

Product-wise Revenue Contribution:

Product Type	Revenue (Rs.)	Share (%)	Cumulative (%)	ABC Class
FRP/GRP Doors	~1,38,00,000	33.7%	33.7%	A
UPVC Windows	~1,10,00,000	26.8%	60.5%	A
Door Frames (FRP)	~97,00,000	23.7%	84.2%	A
Modular Kitchens	~42,00,000	10.2%	94.4%	B
Hardware & Fittings	~23,00,000	5.6%	100%	C

Key Findings — PS-3:

1. Class A Products (84.3% of Revenue): FRP/GRP Doors, UPVC Windows, and Door Frames together contribute 84.3% of total revenue from just 3 out of 5 product categories. These require continuous stock availability and priority procurement.
2. Class B — Modular Kitchens (10.2%): High-unit-value, low-frequency items requiring made-to-order or just-in-time inventory management.
3. Class C — Hardware & Fittings (5.6%): Should have minimal inventory investment with periodic bulk replenishment.

4. Pareto Validation: The analysis validates the 80/20 rule — 60% of products contribute ~84% of revenue.

[Insert Chart: Pareto Chart with Cumulative Line here]

5.4 Purchase Data Analysis

Month	Total Purchase Amount (Rs.)
August 2025	~65,00,000
September 2025	~85,00,000
October 2025	~51,00,000
Total (3 months)	Rs. 2,01,43,597

Purchase spending follows the same cyclical pattern as sales — peaking in September and declining in October — indicating reactive purchasing behavior that can be improved through demand forecasting.

6. Interpretation of Results and Recommendations

(Weightage: 20 marks)

6.1 Interpretation of Key Results

Revenue Cyclicalality (PS-1):

The contrast between September (Rs. 1.90 Cr) and August/October (~Rs. 1.05-1.14 Cr) reflects the natural project procurement cycle in the construction industry. September coincides with active construction seasons, festival-period housing demand, and institutional procurement cycles.

Geographic Concentration (PS-2):

With 62.7% revenue in Bangalore, the company faces market concentration risk. However, the high AOV in Mysuru (Rs. 1,40,485) suggests secondary markets are not just viable but potentially more profitable per transaction. The lower volume is a distribution gap, not a demand gap.

Inventory Imbalance (PS-3):

The ABC analysis confirms that inventory management should shift from a uniform stocking approach to a tiered priority model, preventing stock-outs of Class A items while reducing excess Class C inventory.

6.2 Recommendations

Recommendation 1: Implement Seasonal Demand Planning

Develop a monthly demand forecast model based on historical trends to anticipate the September demand surge and August/October troughs. Use 3-month moving averages and seasonal indices as a starting point; refine with additional historical data.

Recommendation 2: Strategic Market Expansion — Mysuru First

Prioritize Mysuru as the primary expansion market given its highest AOV (Rs. 1,40,485). Establish a local sales representative or distribution point. Hubli should be the secondary priority. Allocate 15-20% of sales effort to these markets over the next 6 months.

Recommendation 3: Government Client Acquisition Drive

Actively pursue government tenders and institutional projects, given 31% higher AOV. Register on GeM (Government e-Marketplace) portal and assign a dedicated team member for tender tracking.

Recommendation 4: ABC-Based Inventory Management

Class A (FRP/GRP Doors, UPVC Windows, Door Frames): Maintain safety stock, set automatic reorder points, and establish priority supplier agreements. Class B (Modular Kitchens): Shift to made-to-order model. Class C (Hardware & Fittings): Periodic bulk procurement with minimal holding.

Recommendation 5: Supplier Relationship Optimization

Negotiate volume-based pricing agreements with top suppliers for Class A raw materials to achieve cost reduction and assured supply during peak demand.

Recommendation 6: Sales Pricing Strategy

Standardize pricing for Builder vs. Government segments with appropriate margin structures. Train sales team to upsell higher-value products to improve per-transaction revenue.

7. Conclusion

This project successfully applied data-driven analytical techniques to address three critical business challenges faced by Ramu Doors Private Limited. Through systematic analysis of 336 sales transactions and comprehensive purchase data spanning August to October 2025, the study revealed:

1. Significant demand cyclicalities with September contributing 46.3% of quarterly revenue, highlighting the need for seasonal demand planning.
2. Geographic concentration risk with Bangalore accounting for 62.7% of revenue, while secondary markets like Mysuru offer higher per-order profitability.
3. Inventory optimization opportunity through ABC classification, where 3 of 5 product categories account for 84.3% of revenue and warrant priority inventory management.

The recommendations provided — seasonal demand planning, strategic Mysuru-first expansion, government client acquisition, ABC-based inventory management, supplier optimization, and pricing strategy refinement — offer a practical, actionable roadmap for sustainable growth. These interventions are expected to improve inventory turnover, reduce stock-out frequency, diversify revenue sources, and enhance the company's profitability and competitive positioning.

8. References

1. Malhotra, N., & Dash, S. (2019). Marketing Research: An Applied Approach. Pearson.
2. Newbold, P. (2013). Statistics for Business and Economics. Pearson.
3. Wirtz, J., & Lovelock, C. (2021). Services Marketing: People, Technology, Strategy. World Scientific.
4. Krajewski, L.J. et al. Operations Management: Processes and Supply Chain. (12th Edition).
5. What is Metadata: [Dataedo - What is Metadata](#)
6. Descriptive Statistics: [Investopedia - Descriptive Statistics](#)

9. Appendix

Appendix A: Proof of Originality of Data

Images of Organisation: [Google Drive - Organisation Images](#)

Video Interaction with Founder: [Google Drive - Video Interaction](#)

Letter from Organisation: [Google Drive - Organisation Letter](#)

Appendix B: Datasets

Purchase Data: [Google Drive - Purchase Data](#)

Sales Data: [Google Sheets - Sales Data](#)